Morten Dall

Contents

Welcome . . . 7
Target audiences . . . 7
Re-use and licensing . . . 7
Contributing . . . 7
How the website is made . . . 7

I Overview

1 Syllabus . . . 11
1.1 Learning outcome and objectives . . . 11

2 Is this for you? . . . 13

3 Schedule . . . 15
3.1 Day 1 . . . 15

4 Learning design . . . 17
4.1 Clear learning goals and a strong narrative . . . 17
4.2 “Code-alongs” as the main activity . . . 17
4.3 Retrieval practice with hands-on exercises . . . 17
4.4 Linking new information with existing knowledge with discussion activities . . . 17
4.5 Minimal use of lectures . . . 17
4.6 Frequent breaks and variation in activities . . . 17
4.7 Easily accessible material that is online, always . . . 18
4.8 A safe and supportive environment . . . 18
4.9 Using origami hats to ask for help . . . 18
4.10 Being involved in the learning process . . . 18

II Pre-workshop

5 Overview . . . 21
5.1 How to read this website . . . 21
5.2 Summary of tasks . . . 21

6 Installing the programs . . . 23
6.1 Windows . . . 23

6.2 MacOS . . . 23
6.3 Linux (Ubuntu) . . . 23

7 Installing R Packages . . . 25

8 Code of conduct . . . 27
8.1 Expected behavior . . . 27
8.2 Unacceptable behavior . . . 27
8.3 Consequences of unacceptable behavior . . . 28

9 Survey . . . 29

III Sessions

10 Introduction to workshop . . . 33
10.1 The Big Picture . . . 33

11 Shiny apps . . . 35
11.1 Learning objectives . . . 35
11.2 The server and the UI . . . 35
11.3 :speech_balloon: Discussion activity: Title . . . 35
11.4 Reading task: ~4 minutes . . . 35
11.5 :book: Adding elements to our UI . . . 36
11.6 BSlib . . . 37
11.7 Summary . . . 40

12 What next . . . 43

Appendices . . . 45

A For teachers . . . 47

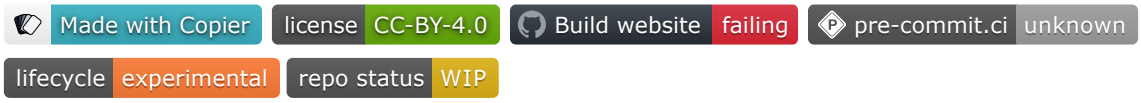
References . . . 49

B Contributing . . . 51
B.1 :bug: Issues and bugs . . . 51
B.2 :pencil2: Adding or modifying content . . . 51
B.3 :file_folder: Support and infrastructure files . . . 51

C Creative Commons Attribution 4.0 International . . . 53
C.1 Creative Commons Attribution 4.0 International Public License . . . 53
C.1.1 Section 1 – Definitions. . . . 53
C.1.2 Section 2 – Scope. . . . 54
C.1.3 Section 3 – License Conditions. . . . 55
C.1.4 Section 4 – Sui Generis Database Rights. . . . 55
C.1.5 Section 5 – Disclaimer of Warranties and Limitation of Liability. . . . 56
C.1.6 Section 6 – Term and Termination. . . . 56

C.1.7 Section 7 – Other Terms and Conditions. 56
C.1.8 Section 8 – Interpretation. 56

Welcome



This shiny workshop is meant as a brief introduction to shiny, arrow, bslib, golem, and plotly. It is designed for doing participatory live-typing, where the teacher demonstrates the tasks on their computer connected to a projector while learners type along on their own computer. The workshop also includes reading tasks, discussion activities, and hands-on exercises.

This website contains all of the material for the workshop, including readings, exercises, presentations, live-typing material, and images. It is structured as a book, with "chapters" as sessions, in order of appearance. We make heavy use of the website throughout the workshop where "type-along" sessions almost identically follow the material on the website (with slight modifications for time or more detailed explanations).

Check out the overview section of the workshop, starting with the Syllabus.

If you plan on attending the workshop, please make sure to complete the Pre-workshop tasks to get set up and ready for the workshop. The pre-workshop tasks include a survey that you need to fill out before the workshop starts.

> Do you find this workshop material useful?
>
> If yes, please consider "starring" our GitHub repository. Starring the repository will save it to your list of saved repositories, so it's easy for you to find again later. As a plus, it helps give our project more visibility :star2:

Target audiences

This website and its content are targeted to three groups:

1. For the **learners** to use during the workshop, both to follow along and also to use as a reference after the workshop ends. A more detailed description of who the learner is can be found in Is this for you?
2. For the **teachers** to use as a guide for when they do the type-along sessions.
3. For those who are **interested in teaching**, who may not have much experience or may not know where to start, to use this website as a guide to running and teaching their own workshops.

Re-use and licensing

The workshop material is licensed under the Creative Commons Attribution 4.0 License, so the material can be used, re-used, and modified, as long as there is attribution to this source. Check out the For teachers section for more details and tips on using this material for teaching.

Contributing

Want to contribute to this workshop? Look through our CONTRIBUTING page for contribution guidelines on how to get started.

How the website is made

The workshop material is created using Quarto to write the material and create the book format, GitHub to host the Git repository of the material, and GitHub Actions with Netlify to build and host the website. The original source material for this workshop is found on the `shiny_workshop` GitHub repository.

I Overview

1 Syllabus 11
1.1 Learning outcome and objectives 11

2 Is this for you? 13

3 Schedule 15
3.1 Day 1 15

4 Learning design 17
4.1 Clear learning goals and a strong narrative 17
4.2 “Code-alongs” as the main activity 17
4.3 Retrieval practice with hands-on exercises 17
4.4 Linking new information with existing knowledge with discussion activities 17
4.5 Minimal use of lectures 17
4.6 Frequent breaks and variation in activities 17
4.7 Easily accessible material that is online, always . 18
4.8 A safe and supportive environment 18
4.9 Using origami hats to ask for help 18
4.10 Being involved in the learning process 18

1. Syllabus

TODO: Include description and overall introduction of workshop

This workshop lasts TODO: Time and is split into the following sessions, listed in the schedule, which will be covered in order:

- Introduction to the workshop
- TODO: Session title
- Concluding remarks

1.1 Learning outcome and objectives

The overall aim of this workshop is to enable you to:

1. Aim one

Broken down into specific objectives for each session, we've designed the workshop to enable you to do the following:

TODO: Session title

1. Objective one

Tangibly, during the workshop you will:

- Learn what a Shiny app is
- Create one with the golem package
- Build a dashboard using the `palmerpenguins` dataset
- Store files as parquet
- Make graphs interactive using plotly

2. Is this for you?

To help manage expectations and develop the material for this workshop, we make a few assumptions about *who you are* as a participant in the workshop.

We assume you can relate to one or more of the following:

- Have intermediate R coding skills
- Are familiar with the tidyverse framework
- Know ggplot2 syntax
- are interested in sharing your data or analysis methods through interactive application

While we have these assumptions to help focus the content of the workshop, if you have an interest in the workshop but don't fit any of the assumptions, *you are still welcome to attend*! We welcome everyone until capacity has been reached.

In addition to the assumptions above, the workshop also has a fairly focused scope, which may also help you decide if this workshop is for you:

- Get a basic introduction to making apps in R
- Learn how to make interfaces with bslib
- Build a modularized app in Shiny
- Learn about the parquet file format
- Make your graphs interactive with plotly

3. Schedule

The workshop is structured as a series of participatory live-coding sessions interspersed with hands-on exercises and group work, using either a practice dataset or some other real-world dataset. There are some lectures given, mainly at the start and end of the workshop. The general schedule outline is shown in the below table. This is not a fixed schedule of the timings of each session—some may be shorter and others may be longer. Instead, it is meant to be an approximate guide and overview.

3.1 Day 1

Time	Session topic

4. Learning design

This workshop is designed using the science of learning. Learning is when new content gets incorporated into long-term storage in the brain in a way that it can be later retrieved. Some of the best ways to enable learning is to do "active learning" by engaging with the material, practicing retrieving what was just covered, repeatedly being exposed to the content, linking the new information with what you already know, and explaining it to someone else.

We use a variety of activities to target the different "pathways" of learning. A big part of learning is being engaged in the *process* of learning, not just learning the content itself. So to help you be engaged in your own learning, we want to involve you in the reasons *why* we do the things we do during this workshop.

Throughout the workshop, we use these practices to support learning.

4.1 Clear learning goals and a strong narrative

The learning goals and the narrative are tightly connected in this workshop. Learning is easier when you have a better idea of where you are going, where you are relative to the end goal, and what to expect along the way.

4.2 "Code-alongs" as the main activity

"Code-alongs" is when the teacher types on their computer that is on the projector and explains what it means as you also type along with them. The connection between using the hands and doing something tangible helps reinforce learning. It also helps the teachers to go slower as we have to both explain and type things out. This helps you learn better. "Less is more" works very well with learning.

4.3 Retrieval practice with hands-on exercises

Hands-on exercises reinforce what was just taught through retrieval practice and repetition. It gives you a chance to engage with the material and the space to try it out on your own. This helps you form a better sense of how to do it yourself.

4.4 Linking new information with existing knowledge with discussion activities

Discussion activities encourage linking what you already know and do in your work with the new information. Because humans are social beings, we also tend to learn better when there is some social aspect to it. For example, explaining something to someone else is one of the best ways to learn, as nicely expressed in the quote "while we teach, we learn" from the Roman philosopher Seneca.

4.5 Minimal use of lectures

Lectures, while extremely common, are forms of information sharing with the lowest rates of learning. In this workshop, we use reading tasks instead, so you can go at your pace, re-read things you may not understand, and take notes, all of which support learning. To help reinforce what you just read, the teacher will briefly summarise and emphasise key points after reading tasks, to repeat the main points and target another part of the brain (the auditory pathway).

4.6 Frequent breaks and variation in activities

Breaks are how your brain can recharge and refocus. The brain is only able to effectively focus on one thing at a time in 30-50 minute blocks of time. That's why we mix together code-alongs, exercises, discussions,

and reading to give your brain some variation. Having an explicit break, even 5 minutes, is enough for the brain to better process and store what was covered.

4.7 Easily accessible material that is online, always

All the material is online on this website, so that it is easy for you to go back to after the workshop. This builds the repetition and retrieval practice part of learning. Our brains also get easily distracted, which is completely normal. So if you get distracted during the workshop, you can simply go to the material and get caught up with where we are. We also make sure to follow the material on the website closely, so that it is easier for you to follow along.

4.8 A safe and supportive environment

Mental safety in a supportive environment is critical for effective learning. It puts the brain into a more relaxed state that allows for optimal processing and ultimately learning. That's why we try to explicitly create that space, for example with the use of our Code of Conduct.

4.9 Using origami hats to ask for help

Asking for help can feel difficult, especially when you are learning and are unsure about something. To make it easier and feel safer, we use origami hats (or stickies) to show how everyone is doing. If you need help, you put up the "help" hat (usually orange) and a helper will come to you as soon as possible. If not, you can put up a "all good" hat (usually blue) to let us know you are following along. Since we often have several helpers, you usually won't wait long. This simple system lets you ask for help without interrupting the teacher or drawing attention, so you can keep following along while you wait. Our goal is to make getting help easier and more comfortable.

4.10 Being involved in the learning process

Now you know a bit more about the thoughts behind how we've designed this workshop to really try to support and enable your own learning! :tada: :grin:

II Pre-workshop

5 Overview 21
5.1 How to read this website 21
5.2 Summary of tasks 21

6 Installing the programs 23
6.1 Windows 23
6.2 MacOS 23
6.3 Linux (Ubuntu) 23

7 Installing R Packages 25

8 Code of conduct 27
8.1 Expected behavior 27
8.2 Unacceptable behavior 27
8.3 Consequences of unacceptable behavior 28

9 Survey 29

5. Overview

In order to participate in this workshop, you must complete everything in this section and finish with **completing the survey (Chapter 9.)** at the end. These tasks are meant to provide everyone with the necessary setup and a shared foundation for the workshop.

Depending on your prior knowledge and skills, these tasks could take between **TIME hrs to finish**, so we suggest planning some time to complete them. While many of the tasks are mostly readings, they are important in making sure you get the most out of the workshop.

5.1 How to read this website

We try to be consistent with the way to present certain information in this website. Specifically, we follow some conventions that you should be aware of:

- Folder names always end with `/`, for example `data/` means the data folder.
- File names always end with their file extension, for example `content.md` means the file is a Markdown file.
- R variables are always shown as is. For instance, for the code `x <- 10`, `x` is a variable because it was assigned with 10.
- Functions always end with `()`, for instance `mean()` or `read_csv()`.
- Sometimes functions have their package name appended with `::` so that you run the code from the specific package, since we likely haven't loaded the package with `library()`. For instance, to install packages from GitHub using the `{pak}` package we use `pak::pak("user/package_name")`. You'll learn about this more later.

5.2 Summary of tasks

Here's a quick overview of the tasks you need to do. Specific details about them are found as you work through the tasks.

If you haven't already, go through the workshop overview section, including the Syllabus, Is this for you?, and Learning design to get an overview of what this workshop contains and the learning design behind the workshop material.

After that, please go through the the pre-workshop tasks designed to establish a shared foundation for the start of the workshop. This includes:

1. Install a version of R, RStudio, Quarto, and Git that is as updated as possible in Chapter 6.. For some people, depending on their institution, this task can take the **longest amount of time** because you have to contact your IT to install these packages.
2. Install necessary R packages.
3. TODO: Add more tasks
4. Read the syllabus (Chapter 1.).
5. Read the Code of Conduct (Chapter 8.).
6. Complete the pre-workshop survey (Chapter 9.): This survey is pretty quick, maybe ~10 minutes. There will be some bits from the tasks that you will need to paste into this survey.

Check each section for exact details on completing these tasks.

> Tip
>
> You will be taken through the pre-workshop tasks in order by clicking the arrow at the bottom of each page :arrow_right:

6. Installing the programs

In order to participate in the workshop, you'll need to install these programs. You may already have some of them installed and if you did, please make sure that they are at least the minimum versions listed below. If not, you will need to update them.

6.1 Windows

1. R: Any version above . If you have used R before, you can confirm the version by running `R.version.string` in the Console.
2. RStudio: Any version above . If you have installed it before, check the current version by going to the menu "Help -> About RStudio".
3. Git: Select the "Click here to download" link. Git is used throughout many sessions in the workshop. When installing, it will ask for a selecting a "Text Editor" and while we won't be using this in the workshop, Git needs to know this information so choose Notepad.

6.2 MacOS

1. R: Any version above . If you have used R before, you can confirm the version by running `R.version.string` in the Console. If you use Homebrew (which we strongly recommend installing and using), installing R is as easy as opening a Terminal and running:

```
brew install --cask r
```

2. RStudio: Any version above . If you have installed it before, check the current version by going to the menu "Help -> About RStudio". With Homebrew:

```
brew install --cask rstudio
```

3. Git: Git is used throughout many sessions in the workshop. With Homebrew:

```
brew install git
```

6.3 Linux (Ubuntu)

1. R: Any version above . If you have used R before, you can confirm the version by running `R.version.string` in the Console.

```
sudo apt -y install r-base
```

2. RStudio: Any version above . If you have installed it before, check the current version by going to the menu "Help -> About RStudio".
3. Git: Git is used throughout many sessions in the workshop.

```
sudo apt install git
```

All these programs are required for the workshop, even Git. Git, which is a software program to formally manage versions of files, is used because of it's popularity and the amount of documentation available for it. Check out the online book Happy Git with R, especially the "Why Git" section, for an understanding

on why we are teaching Git. Windows users tend to have more trouble with installing Git than macOS or Linux users. See the section on Installing Git for Windows for help.

> **i** Note
>
> Some pictures may show a Git pane in RStudio, but you may not see it. If you haven't created or opened an RStudio R Project (which is taught in the introductory workshop), the Git pane *does not* show up. It only shows up in R Projects that use Git to track file changes.

> **i** Note
>
> *A note to those who have or use work laptops with restrictive administrative privileges*: You may encounter problems installing software due to administrative reasons (e.g. you don't have permission to install things). Even if you have issues installing or updating the latest version of R or RStudio, you will likely be able to continue with the workshop as long as you have the minimum version listed above for R and for RStudio. If you have versions of R and RStudio that are older than that, you may need to ask your IT department to update your software if you can't do this yourself. Unfortunately, Git is not a commonly used software for some organizations, so you may not have it installed and you *will* need to ask IT to install it. We *require* it for the workshop, so please make sure to give IT enough time to be able to install it for you prior to the workshop.

Once R, RStudio, and Git have been installed, open RStudio. *If you encounter any troubles during these pre-workshop tasks*, try as best as you can to complete the task and then let us know about the issues in the pre-workshop survey. If you continue having problems, indicate on the survey that you need help and we can try to book a quick video call to fix the problem. Otherwise, you can come to the workshop 15-20 minutes earlier to get help.

If you're unable to complete the setup procedure due to unfixable technical issues, you can use Posit Cloud (to use RStudio on the cloud) as a final solution in order to participate in the workshop. For help setting up Posit Cloud for this workshop, refer to the Posit Cloud setup guide.

7. Installing R Packages

We will be using specific R packages for the workshop, so you will need to install them. To make it easier, we've created a custom R package called `r3` that will install all the necessary packages for you for this workshop. Run these two functions in the Console by copying and pasting them to install our custom `r3` package as well as the `{tidyverse}` package.

```
install.packages("r3", repos = c("https://rostools.r-universe.dev", "https://cloud.r-project.org"))
install.packages("tidyverse")
```

Tip

Installing packages is tricky, especially on some work laptops, and doesn't always work as expected. Here are some other methods to try if you have trouble installing the packages with the code above.

For Windows users, you may have to install RTools to use either the `{pak}` or `{remotes}` method. The version of RTools you install must match the version of R that you have installed. For example, if you have R version 4.4, you need to install RTools 4.4.

First, try installing the `r3` package directly from GitHub with `{pak}`:

```
install.packages("pak")
pak::pak("rostools/r3")
```

If that doesn't work, try with `{remotes}`:

```
install.packages("remotes")
remotes::install_github("rostools/r3")
```

If you encounter any errors with either of these functions, try to restart R by going to "Sessions -> Restart R" and re-run them again. If that also doesn't work, try to complete the other tasks as best you can, complete the survey, and let us know you have a problem in the survey. We'll email you at least a week before the workshop starts to try to help you get it working.

Tip

When you see a command like `something::something()`, for example with `pak::pak()`, you would "read" this as:

> R, can you please use the `pak()` function from the `{pak}` package.

The normal way of doing this would be to load the package with `library(pak)` and then running the command (`pak()`). But by using the `::`, we tell R to directly use a function from a package, without needing to load the package and all of its other functions too. We use this trick because we only want to use the `pak()` command from the `{pak}` package and not have to load all the other functions as well. In this workshop we will be using `::` often.

8. Code of conduct

Learning happens when we feel safe and supported. Since we want you to learn during this workshop, we want you to feel safe and supported. We also want all learners to have an enjoyable and fulfilling experience while participating in this workshop. So we expect all learners to show respect, kindness, and courtesy to other learners throughout the whole workshop and to help create a positive and safe learning environment.

While most people and (sub-)cultures have an implicit, assumed set of practices and expectations for behaviour, we believe strongly in effective and clear communication. This includes making what is normally implicit expectations to instead be explicit expectations.

Teachers and helpers will uphold and embody this code in all spaces and activities of the workshop, and we expect you as the learner to do the same. In fact, by participating in this workshop, you indicate that you agree to follow these explicit expectations. We have this code of conduct to communicate to you that we really value and want to nurture a space that encourages and enhances learning. Otherwise, why would we have a workshop if you can't or aren't able to effectively learn anything from it! :zany_face:

8.1 Expected behavior

We want a welcoming, supportive, safe, and fun environment not only for ourselves as teachers and helpers, but also for all learners, regardless of background and identity. This means that we do not tolerate behaviour that is disrespectful to any of our teachers, helpers, or learners (see below for unacceptable behaviour).

We encourage the following kinds of behaviours in all interactions during this workshop that will foster a positive and professional learning environment:

- Show kindness, courtesy, and respect towards other workshop members.
- All interactions should be professional regardless of platform: either online or in-person.
- Be aware of your words by using welcoming and inclusive language. Words matter, a lot!
- Be respectful to different viewpoints and experiences.
- Gracefully accept constructive criticism.
- Focus on what is best for everyone in the workshop.

8.2 Unacceptable behavior

While we assume you will not do these behaviours, we want to clearly communicate what unacceptable behaviour is. It is any behaviour, as actions or words, that excludes, intimidates, or causes discomfort to others. We do not tolerate discrimination or harassment based on characteristics that include, though are not limited to, gender identity and expression, sexual orientation, disability, physical appearance, body size, citizenship, nationality, ethnic or social origin, pregnancy, familial status, veteran status, genetic information, religion or belief (or lack thereof), membership of a national minority, property, age, education, socio-economic status, technical choices, and experience level. We of course also don't tolerate verbally or physically violent behaviour towards others.

Examples of unacceptable behavior includes:

- Written or verbal comments that exclude people on the basis of membership of any specific group.
- The display of sexual or violent images.
- Unwelcome sexual attention, advances, and/or physical contact.
- Sustained disruption of workshop sessions or communications.
- Insults or put downs.
- Sexist, racist, homophobic, transphobic, ableist, or exclusionary jokes.

8.3 Consequences of unacceptable behavior

Learners who are asked, either in person or online, to stop any inappropriate behavior are expected to comply immediately. If a learners engages in behaviour that violates this code of conduct, the organizers, teachers, or helpers may take any action they deem appropriate, including warning the offender or removing them from the workshop.

If you feel in any way unsafe or mistreated, you may contact us privately—all communication is confidential.

Thank you for helping make this a welcoming, friendly workshop for all :smile:

> **i** Note
>
> *Acknowledgements*: Large portions of this code of conduct were taken and modified from other codes of conduct, specifically Software Carpentry and UofTCoders.

9. Survey

We have a Code of Conduct. If you haven't read it, **read it now**. The survey involves a question about Conduct. We want to make sure this workshop is a supportive and safe environment for learning, so this Code of Conduct is quite important.

If you haven't read the sections under "Workshop overview" and "Pre-workshop tasks", **please read them now**.

You're almost done. Please fill out the pre-workshop survey to finish this section:

Pre-workshop survey :tada:

See you at the workshop! :wave:

III Sessions

10 Introduction to workshop 33
10.1 The Big Picture 33

11 Shiny apps 35
11.1 Learning objectives 35
11.2 The server and the UI 35
11.3 :speech_balloon: Discussion activity: Title 35
11.4 Reading task: ~4 minutes 35
11.5 :book: Adding elements to our UI 36
11.6 BSlib 37
11.7 Summary 40

12 What next 43

10. Introduction to workshop

Introduction slides

> **i** Teacher note
>
> The slides contain speaking notes that you can view by pressing 'S' on the keyboard.

10.1 The Big Picture

TODO: Add text highlighting the bigger picture of why we teach this.

11. Shiny apps

To build a shiny app we must first understand how it works. A shiny app consists of two core components: a server and a user interface (UI). In this section, we will explain how they are connected and work together to create the shiny app.

11.1 Learning objectives

The learning objectives for this session are:

1. Objective one

11.2 The server and the UI

A shiny app can take a static dataset and make it available for the user to explore. Depending on how you build the app, you can allow the user to search for specific search terms, compare expression levels, or explore interactive figures to find new genes affected in your dataset.

11.3 :speech_balloon: Discussion activity: Title

Time: \3 minutes.

Discuss with the person next to you - what kind of data do you have that could benefit from being interactive? Do you have an idea for an app you want to make?

11.4 Reading task: ~4 minutes

A shiny app can get incredibly complex and consist of many interconnected scripts. But at it's simplest, a shiny app consists of two elements that work together.

At it's most simple, a shiny app can be one simple function. The following code will create a shiny app (although an empty one):

```
app <- shiny::shinyApp(
  ui = shiny::fluidPage(),
  server = function(input, output){}
)
shiny::runApp(app)
```

The UI (short for user interface) is what we see on the screen. The code contains information about what is shown on the page (such as plots, buttons and images) and in what layout (organized in columns, with a navigation bar, etc.).

The server contains information about the objects in the UI, and how the app should respond to user input. It contains information about what data is displayed in the plot, what happens when you press a button, and how the app should update when specific events happen.

A major part of building a shiny app is understanding the interplay between the UI and the server.

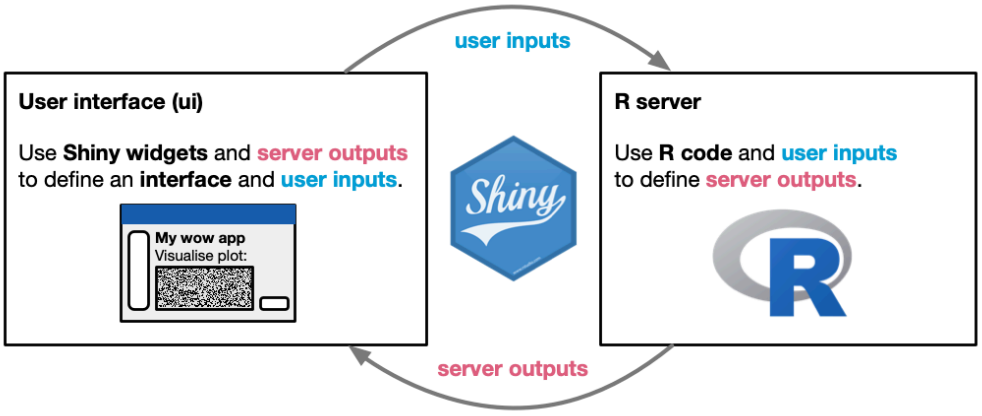


Figure 11.1: A graphical representation of the interplay between the UI and the server. From introduction to data science

11.5 :book: Adding elements to our UI

Time: \6 minutes.

Shiny comes with a lot of options for building a user interface. Often you can add plots directly into a Shiny interface. lets say we want to make a graph from the classical `iris` data set and add it into our empty app above.

```
iris_plot <- ggplot2::ggplot(iris,
                             ggplot2::aes(x = Sepal.Length, y = Petal.Length, color =
Species)) +
  ggplot2::geom_point()

iris_plot
```

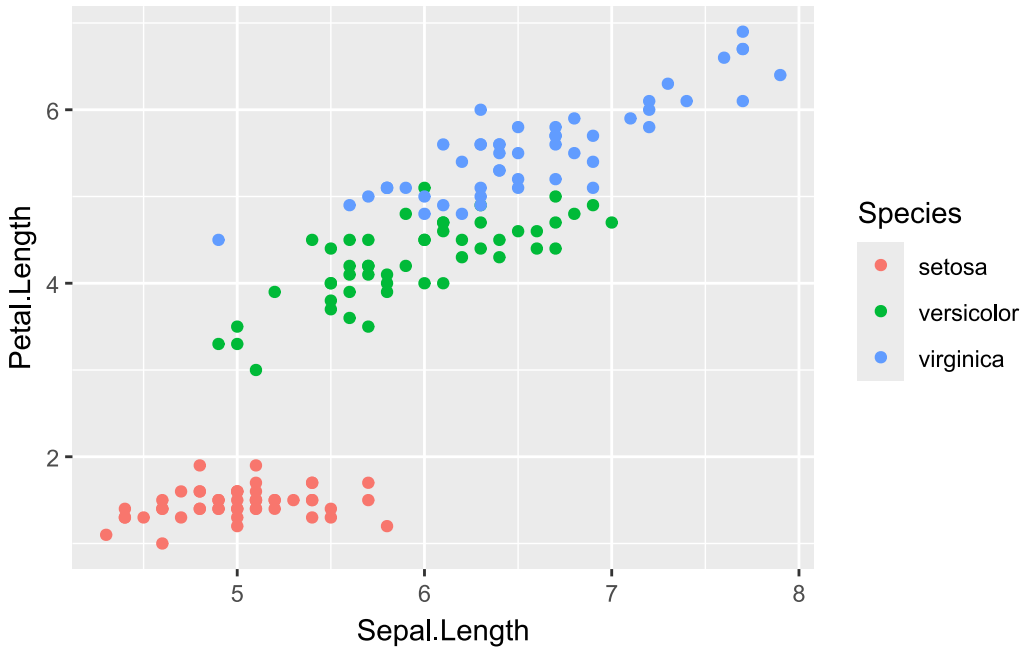


Figure 11.2: Our classical iris plot

To add this to our app, we need to tell Shiny two things: that we want a page in our page layout, and what we want the plot to contain.

```
app <- shiny::shinyApp(
  ui = shiny::fluidPage(
    shiny::plotOutput(outputId = "iris_plot"
    )
  ),
  server = function(input, output){
    output$iris_plot <- shiny::renderPlot(
      iris_plot
    )
  }
)
shiny::runApp(app)
```

If we take the code above and paste it into the console, we should get an app with the plot. We have now created content for our app!

11.6 BSlib

It is entirely possible to build a shiny app out of components from the shiny package itself. However, the "modern" shiny design focuses on a different package called "bslib" which provides access to the highly flexible bootstrap framework. Bslib allows a number of UI components with a high degree of customization, and which renders nicely, even on a small screen such as a phone.

Bslib features both pages (used to define the overall rules for our app page) and the components of the pages (especially as units of UI elements called "cards"). We can take our previous code and modify it to become a navbar page.

```
app <- shiny::shinyApp(
  ui = bslib::page_sidebar(
    title = "Our Iris plot",
    sidebar = "here is our sidebar",
    shiny::plotOutput(outputId = "iris_plot"
    )
  ),
  server = function(input, output){
    output$iris_plot <- shiny::renderPlot(
      iris_plot
    )
  }
)
shiny::runApp(app)
```

If we paste this code into our console and runs it (rerunning the code for the iris plot if necessary), we will see we now have a sidebar where we can put something in. Let's leave it empty for now, and instead focus on the other functionality of bslib we teased before - the cards.

Cards are an effective way to organize plots and other UI components. They are easy to implement - we simply wrap our UI content (in this case the Iris plot) in `bslib::card()`

```
app <- shiny::shinyApp(
  ui = bslib::page_sidebar(
    title = "Our Iris plot",
    sidebar = "here is our sidebar",
    bslib::c(shiny::plotOutput(outputId = "iris_plot")
    )
  ),
  server = function(input, output){
    output$iris_plot <- shiny::renderPlot(
      iris_plot
    )
```

```
  }
)
shiny::runApp(app)
```

In itself this does not do much. However, cards have some attributes such as headers that can be modified if we for example want to give our plot a header.

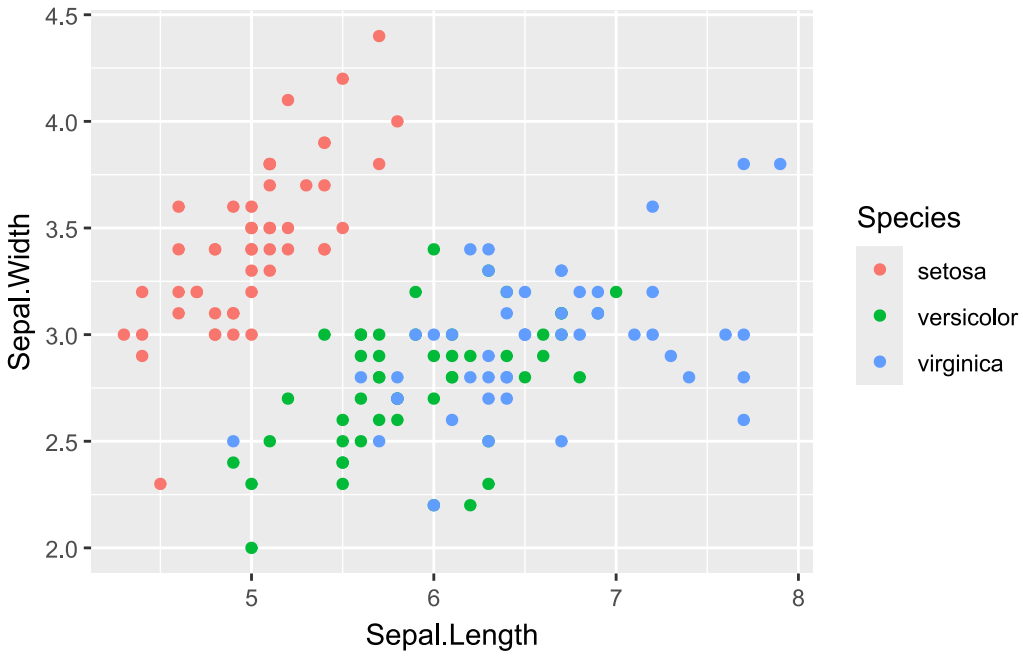
```
app <- shiny::shinyApp(
  ui = bslib::page_sidebar(
    title = "Our Iris plot",
    sidebar = "here is our sidebar",
    bslib::card(
      bslib::card_header("A beautiful plot about flowers"),
      shiny::plotOutput(outputId = "iris_plot")
    )
  ),
  server = function(input, output){
    output$iris_plot <- shiny::renderPlot(
      iris_plot
    )
  }
)
shiny::runApp(app)
```

The true power of cards becomes evident once we start having several of them. Let's make a few more plots so we can see how bslib behaves once we have multiple:

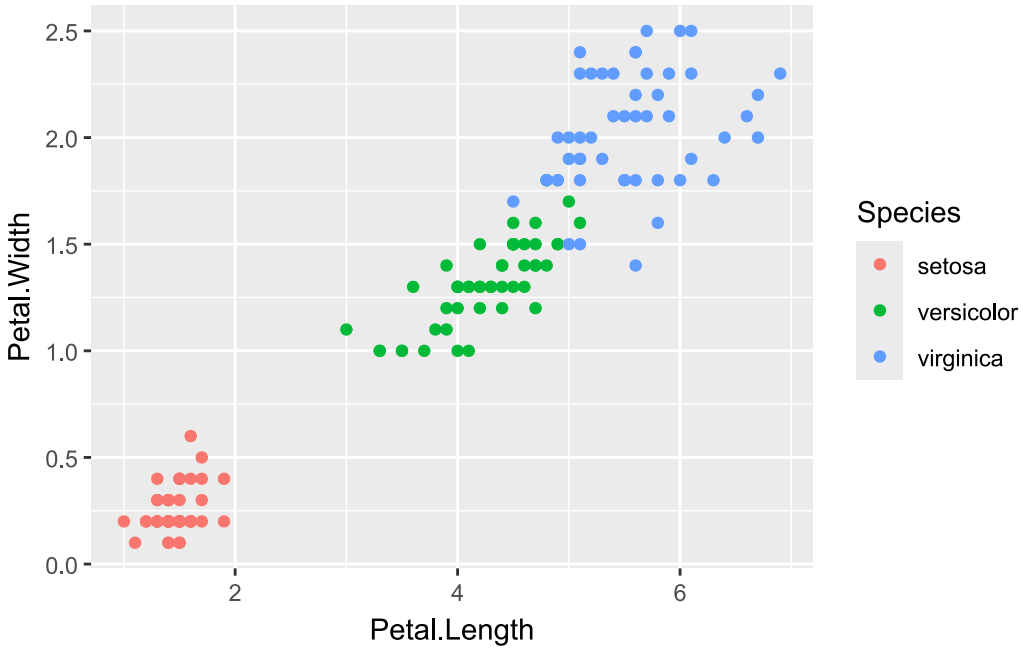
```
iris_plot_2 <- ggplot2::ggplot(iris,
                               ggplot2::aes(x = Sepal.Length, y = Sepal.Width, color
= Species)) +
  ggplot2::geom_point()


iris_plot_3 <- ggplot2::ggplot(iris,
                               ggplot2::aes(x = Petal.Length, y = Petal.Width, color
= Species)) +
  ggplot2::geom_point()

iris_plot_2
```




```
iris_plot_3
```



Lets add our two new plots to our layout, copy it to the console, and run it:

```
app <- shiny::shinyApp(
  ui = bslib::page_sidebar(
    title = "Our Iris plot",
    sidebar = "here is our sidebar",
    bslib::card(
      bslib::card_header("A beautiful plot about flowers"),
      shiny::plotOutput(outputId = "iris_plot")
    ),
    bslib::card(
      bslib::card_header("Another beautiful plot about flowers"),
      shiny::plotOutput(outputId = "iris_plot_2")
```

```
    ),
    bslib::card(
      bslib::card_header("OKay I think I've had enough flower plots for now"),
      shiny::plotOutput(outputId = "iris_plot_3")
    )
  ),
  server = function(input, output){
    output$iris_plot <- shiny::renderPlot(
      iris_plot
    )
    output$iris_plot_2 <- shiny::renderPlot(
      iris_plot_2
    )
    output$iris_plot_3 <- shiny::renderPlot(
      iris_plot_3
    )
  }
)
shiny::runApp(app)
```

It looks nice! if we want to change the order of our cards, we can add another layer from bslib, and define how many columns we want. We wrap all our cards in a function called `bslib::layout_column_wrap()` and add the argument `width  = 1/2` to get 2 columns (if this is indeed what we want). If we instead had written `width = 1/3` we would get 3 columns and so forth.

```
app <- shiny::shinyApp(
  ui = bslib::page_sidebar(
    title = "Our Iris plot",
    sidebar = "here is our sidebar",
    bslib::layout_column_wrap(
      width = 1/2,
      bslib::card(
        bslib::card_header("A beautiful plot about flowers"),
        shiny::plotOutput(outputId = "iris_plot")
      ),
      bslib::card(
        bslib::card_header("A beautiful plot about flowers"),
        shiny::plotOutput(outputId = "iris_plot_2")
      ),
      bslib::card(
        bslib::card_header("A beautiful plot about flowers"),
        shiny::plotOutput(outputId = "iris_plot_3")
      )
    )
  ),
  server = function(input, output) {
    output$iris_plot <- shiny::renderPlot(iris_plot)
    output$iris_plot_2 <- shiny::renderPlot(iris_plot_2)
    output$iris_plot_3 <- shiny::renderPlot(iris_plot_3)
  }
)
shiny::runApp(app)
```

There is much more functionality to building a good user interface and there are many resources on bslib. But for now we know how to build a Shiny interface

11.7 Summary

In this session we:

- Learned about UI and server
- Explored the relationship between outputs and render functions

- Introduced bslib and showed how it can be used to make pages and UI elements

Next session, we will show how we can build a shiny app so we have a place for our UI

12. What next

Slides

> **Teacher note**
>
> The slides contain speaking notes that you can view by pressing ‘S’ on the keyboard.

Appendices

A. For teachers

A more general guide to teaching or helping is found on the Guides website. This page has information relevant specifically to this workshop.

- For created teams, assign one teacher or helper to each group to help them specifically during the final group project.

References

B. Contributing

Check out the Contributing Guide for more details on how to contribute. For guidelines on how to be a helper or teacher, check out the For teachers page. As always, make sure to adhere to the Code of Conduct.

B.1 :bug: Issues and bugs

The easiest way to contribute is to report issues or bugs that you might find while using or reading the workshop material. You can do this by creating a new issue on our GitHub repository.

B.2 :pencil2: Adding or modifying content

To contribute to the workshop, you first need to install uv and justfile. We use uv and justfile to manage our project, such as to run checks and build the workshop website. Both the uv and justfile websites have a more detailed guide on using uv, but below are some simple instructions to get you started.

It's easiest to install uv first and then install justfile using uv. Then, to install justfile, run:

```
uv tool install rust-just
```

We keep all our development workflows in the `justfile`, so you can explore it to see what commands are available. To see a list of commands available, run:

```
just
```

As you contribute, make sure your changes will pass our checks by opening a terminal so that the working directory is the root of this workshop's parent/root folder and running:

```
just run-all
```

When committing changes, please try to follow Conventional Commits as Git messages. Using this convention allows us to be able to automatically create a release based on the commit message by using Cocogitto. If you don't use Conventional Commits when making a commit, we will revise the pull request title to follow that format. That's because we use squash merges when merging pull requests, so all other commits in the pull request will be squashed into one commit.

B.3 :file_folder: Support and infrastructure files

- `.copier-answers.yml`: Contains the answers you gave when copying the project from the template. **You should not modify this file directly.**
- `.gitignore`: This ignore file tells Git which files to not track. Unless you know what you are doing, it's best to not touch this file.
- `_quarto.yml`: Quarto configuration file for the website, including settings for the website, such as the theme, navigation, and other options.
- `_metadata.yml`: Quarto metadata file for the website, including information about the project, such as the titles and GitHub names.
- `.pre-commit-config.yaml`: Pre-commit configuration file for managing and running checks before each commit.
- `.config/`: Contains configuration files for various tools used in the project, such as:
  - `typos.toml`: typos spell checker configuration file.
  - `rumdl.toml` and `panache.toml`: rumdl and Panache configuration file for formatting Markdown files in the project.
- `.zenodo.json`: Structured citation metadata for your project when archived on Zenodo. This is used to add the metadata to Zenodo when a GitHub release has been uploaded to Zenodo.
- `justfile`: `just` configuration file for scripting project tasks.

- `.editorconfig`: Editor configuration file for EditorConfig to maintain consistent coding styles across different editors and IDEs.
- `CONTRIBUTING.md`: Guidelines for contributing to the project.
- `.github/`: Contains GitHub-specific files, such as issue and pull request templates, workflows, dependabot configuration, pull request templates, and a CODEOWNERS file.

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